

Celecoxib (CELEBREX®) Adjunctive Therapy of Familial Adenomatous Polyposis (FAP)

Subpart H Approval

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ODAC

March 12-13, 2003



ODAC Meeting

Pharmacia Attendees

- **Langdon Miller, MD**
Vice President, Clinical Research
- **Kenneth Verburg, MD**
Vice President, Clinical Research
- **P.K. Narang, PhD**
Senior Director, Regulatory Affairs
- **Kerry Barker, PhD**
Director, Biostatistics
- **Bernard Levin, MD (consultant)**
UT MD Anderson Cancer Center
- **Patrick Lynch, MD (consultant)**
UT MD Anderson Cancer Center

Summary

Pharmacia is committed to fulfilling Subpart H requirements

- **Successful completion of ZINECARD[®],
CAMPTOSAR[®] commitments**
- **CELEBREX FAP post-approval program
underway**

Presentation Agenda

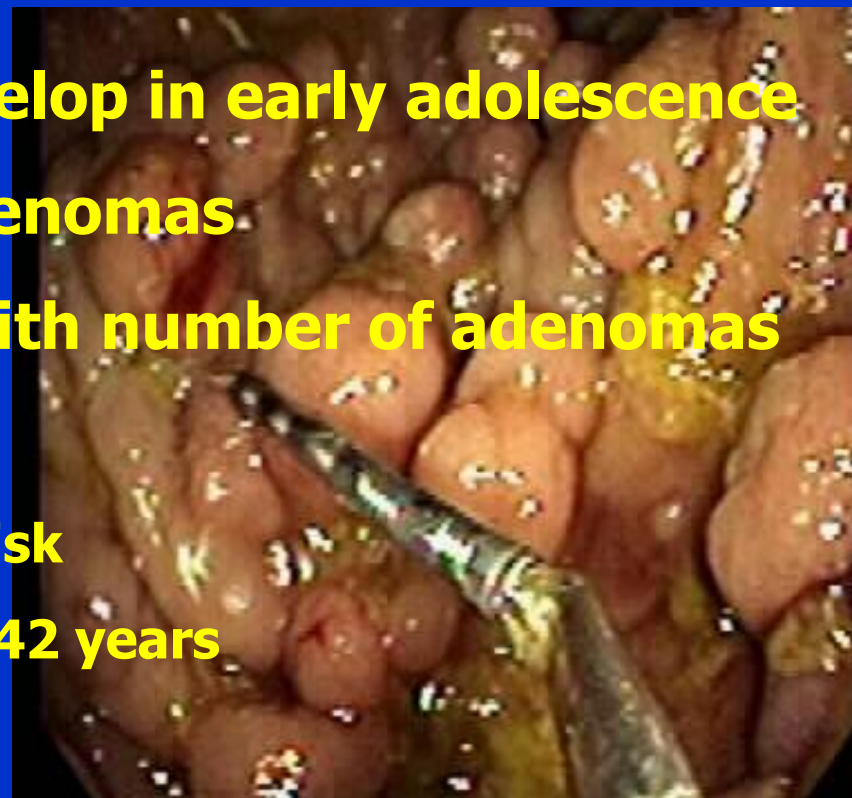
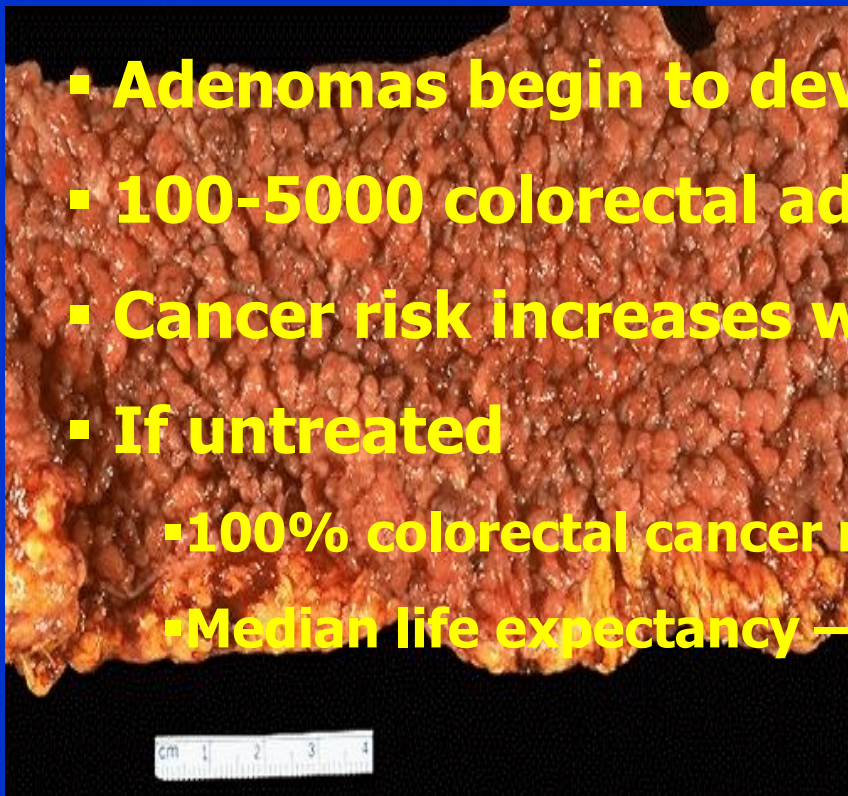
- **FAP overview**
- **Basis for celecoxib approval**
- **Indication**
- **Subpart H commitments**
- **Conclusions**

FAP Overview

- **Rare, life-threatening disease**
- **Autosomal dominant inheritance**
 - Germline APC mutations (5q21)
- **~ 300 new patients/year in US**
- **Accounts for 1% of all colorectal cancers**

Natural History

- Adenomas begin to develop in early adolescence
- 100-5000 colorectal adenomas
- Cancer risk increases with number of adenomas
- If untreated
 - 100% colorectal cancer risk
 - Median life expectancy – 42 years



Disease Management

- **Lifetime endoscopic surveillance**
- **Initial colon resection 18-20 years of age**
- **Repeated surgeries**

- **Interest in developing medical treatment as an adjunct to surgery**

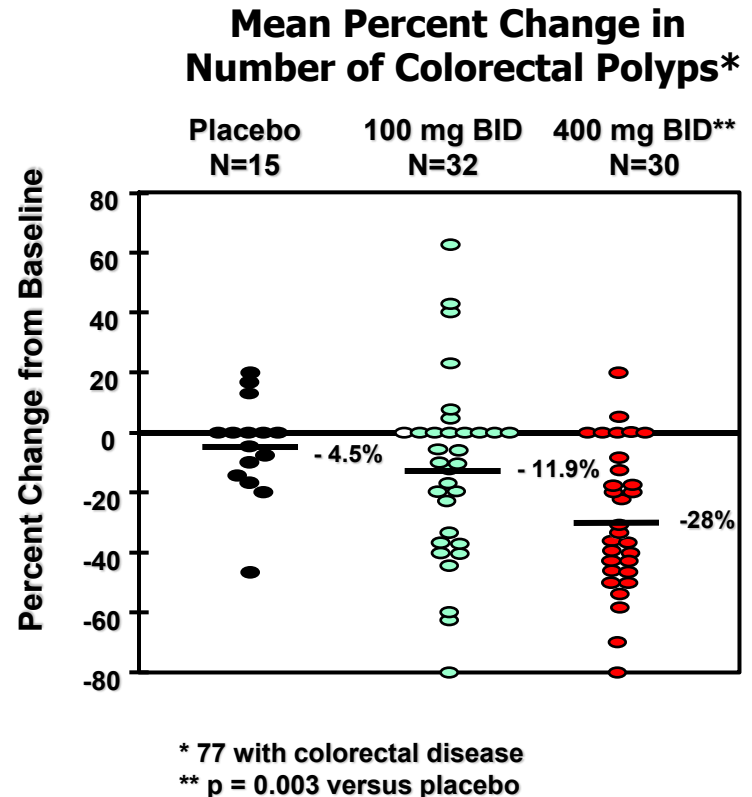
Pivotal Registration Trial

Basis for Approval

Description:	Double-blind, placebo-controlled study of celecoxib in patients with FAP
Sites:	U.T. M.D. Anderson, St. Mark's (UK)
Treatment Groups:	Placebo Celecoxib (100, 400 mg po BID)
Primary Endpoint:	Percent change in the number of colorectal adenomas
Duration of Therapy:	6 months

Results of Pivotal Trial

- **Largest prospective, randomized trial conducted in FAP**
 - 2 years to complete
 - 83 patients
- **Efficacy: 400 mg BID**
 - 28% reduction in mean polyp number compared to baseline
 - Secondary endpoints confirmatory
- **Safety: 400 mg BID**
 - Well tolerated



FAP Indication

- **To reduce the number of adenomatous colorectal polyps in familial adenomatous polyposis (FAP), as an adjunct to usual care (e.g., endoscopic surveillance, surgery)**
- **It is not known whether there is a clinical benefit from a reduction in the number of colorectal polyps in FAP patients.**
- **It is not known whether the effects of CELEBREX treatment will persist after CELEBREX is discontinued**
- **The efficacy and safety of CELEBREX treatment in patients with FAP beyond six months have not been studied**

Subpart H Commitments

- **FAP phenotype suppression study**
 - **Designed to verify clinical benefit**
 - **Placebo-controlled trial in patients who are genotypically positive (have APC mutation) but phenotypically negative (have not yet developed polyps)**
- **FAP registry**
 - **Determine both efficacy and safety parameters associated with short and long-term exposure**

Subpart H Commitments

- **FAP phenotype suppression study**
 - **Placebo-controlled trial in patients who are genotypically positive (have APC mutation) but phenotypically negative (have not yet developed polyps)**
- **FAP registry**
 - **A long-term registry of clinical outcomes**

Phenotype Suppression Study

Proposed Design

Description:	Phase III study of celecoxib in genotype-positive, phenotype negative children with FAP
Treatment groups:	Placebo Celecoxib (400 mg po BID) 1:2 randomization
Sample size:	N = 231
Duration of therapy:	5 years
Primary endpoint:	Time to first adenoma

Phenotype Suppression Study

Brief Chronology of Events

12/99

- **FDA agrees with study concept**

4/00

- **NCI/Pharmacia collaboration**
 - NCI issues request for proposals (RFP)
 - Pharmacia to provide drug and monetary support

7/00

- **RFP awarded (8 collaborating institutions)**
 - MD Anderson - lead institution
 - Creighton University
 - Memorial Sloan-Kettering Cancer Center
 - Cleveland Clinic
 - Texas Children's Hospital
 - University of California San Francisco
 - Mt Sinai Hospital (Toronto)
 - St Mark's Hospital (England)

Phenotype Suppression Study

Brief Chronology of Events

8/00

- **Study concerns among collaborators**
 - pediatric population
 - celecoxib dose not established in children
 - pilot dose-ranging trial needed

10/00

- **Draft phase I protocol developed**
 - submitted to NCI and Pharmacia

1/01

- **Phase I/III program submitted to FDA**

4/01

- **FDA accepts program**

2/01-12/01

- **3 protocol revisions required**

1/02

- **Protocol approved by NCI**

Phenotype Suppression Study

Phase I Design

Description:	Phase I study of celecoxib in genotype-positive children with FAP
Sites:	U.T. M.D. Anderson, Texas Children's Hospital, Cleveland Clinic
Design:	Dose escalation trial in successive cohorts of 6 patients
Treatment groups:	Placebo Celecoxib (2, 4, 8 mg/kg po BID)
Sample size:	N = 18
Duration of therapy:	3 months for each cohort
Primary endpoint:	Safe dose in children

Phenotype Suppression Studies Brief Chronology of Events

2/02

- **MDACC IRB approval**

5/02

- **Final phase I protocol submitted to FDA**

6/02

- **Site initiation meeting held**

6/02

- **Development delays with investigational 50-mg orally dispersible tablet**

8/02

- **Protocol revised to use commercial capsule formulation**

12/02

- **First patient enrolled in phase I study**
 - **Current accrual 6 of 18 (first cohort)**

1Q04

- **Phase III trial**
 - **Last patient in 2006, final report 2011**

Subpart H Commitments

- FAP phenotype suppression study
 - Placebo-controlled trial in patients who are genotypically positive (have APC mutation) but phenotypically negative (have not yet developed polyps)
- **FAP registry**
 - **A long-term registry of clinical outcomes**

FAP Registry Initial Design

Description:

Observational

Patient Population:

**Patients receiving celecoxib
Historical controls**

Primary Endpoints:

**Time to FAP-related events
Adverse events**

FAP Registry

Brief Chronology of Events

12/99

- **FDA agrees with concept**

2-4/00

- **Concerns raised in discussions with experts**
 - **Patients who would receive drug in clinical practice not yet characterized**
 - **Changes in clinical management might confound comparison**
 - **Complexity of surgical decisions would introduce variability**
 - **Time to FAP-related events is often long**

5/00

- **Alternative to registry explored**
 - **Collaboration with NCI and Ilex Pharmaceuticals**
 - **Combination trial celecoxib $\pm$ difluoromethylornithine (DFMO)**

12/00

- **Alternative proposal submitted to FDA**

FAP Registry Brief Chronology of Events

4/01

- **FDA feedback**
 - **Proposed DFMO study did not address Subpart H commitments**
 - **FDA still considered a registry worthwhile**
 - **Acknowledged that new therapies and differences in clinical practice may confound analysis**
- **Efforts refocused on FAP registry**

FAP Registry Brief Chronology of Events

5/01

- **Partnership pursued with Collaborative Group of the Americas on Inherited Colorectal Cancer (CGA)**
 - Consortium of 17 registries and clinics in US, Canada and South America

10/01

- **Concept for provider-driven registry presented at CGA annual meeting by MDACC**

11/01 - 3/02

- **Web-based registry utilizing CGA centers designed and developed by MDACC**

4/02

- **Full web-based protocol submitted to CGA membership**

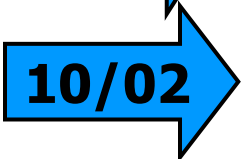
7/02

- **On further review, CGA members express lack of enthusiasm for registry**
 - Too labor-intensive

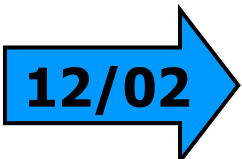
FAP Registry Brief Chronology of Events

**7/02**

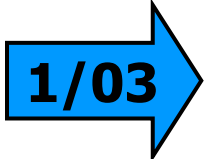
- **MDACC revises registry**
 - Patients to enter own data via internet

**10/02**

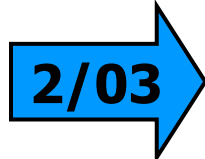
- **CGA annual meeting**
 - Revised proposal presented

**12/02**

- **Prototype of patient-driven internet registry developed at MDACC**
 - Protocol submitted to MDACC IRB

**1/03**

- **MDACC IRB does not recommend approval**
 - Lack of source data verification
 - Patient confidentiality issues

**2/03**

- **Revised protocol with established registries developed**
 - Protocol summary submitted to FDA

FAP Registry Proposed Design

Description:	Observational
Patient Population:	Patients receiving celecoxib Historical controls
Sites under consideration:	Established FAP registries
Objectives:	- Describe characteristics of patients who receive celecoxib in clinical practice - Describe patterns of celecoxib use in disease management - Evaluate long-term safety of celecoxib - Assess whether celecoxib use may alter management of FAP - Determine impact on incidence of FAP-related events (eg, polypectomy, surgery, cancer, desmoids, death)

Conclusions

Pharmacia is committed to fulfilling Subpart H requirements

- **Phenotype suppression program to verify clinical benefit has begun**
- **Continuing progress in implementing a revised FAP registry**